Saeed Ahmad

Machine Learning Engineer

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SUMMARY

Machine Learning Engineer with 3+ years of industry experience spanning deep learning research, LLM systems, agentic AI, medical imaging, model fine-tuning, and production infrastructure. Currently building a role-aware enterprise HR chatbot (Microsoft Fabric → Neo4j with source-system RBAC → LangGraph, on OpenShift); trained diabetic retinopathy lesion detection models on UWF and fundus imaging. Achieved **TEDS 98.51 on PubTabNet** and **99.45 on FinTabNet**; fine-tuned LLaVA 7B with PEFT; built multi-agent systems processing **1M+ records across 20 languages**; reduced inference latency **5**× via TensorRT, OpenVINO, and PagedAttention. MS in Artificial Intelligence (CGPA 3.85), paper under review.

SKILLS

- **AI / ML:** Deep Learning, LLMs, Agentic AI, Multi-Agent Systems, RAG, Computer Vision, NLP, Multimodal Learning, Self-Supervised Learning, Model Fine-Tuning, Knowledge Distillation, GANs, Transformers, Document AI, Table Structure Recognition, OCR, Entity Linking, Knowledge Graphs, Graph Ontology Design, Medical Imaging, Lesion Detection
- **Fine-Tuning:** LoRA, QLoRA, PEFT, Full Fine-Tuning, PyTorch DDP (2–4 GPUs)
- **Optimization:** ONNX, TensorRT, OpenVINO, PagedAttention, Dynamic Batching, Quantization
- **Frameworks:** PyTorch, TensorFlow, LangGraph, LangChain, Hugging Face Transformers, FastAPI, Ray, gRPC
- **Languages:** Python, C++, Go, SQL
- **Infra / Cloud:** Microsoft Azure, Azure OpenAI, Red Hat OpenShift, AWS (Lambda, EC2), RunPod, Lambda Labs, DigitalOcean, GCP, Docker, NVIDIA GPU, CI/CD, GitHub Actions, CUDA
- **Data:** Microsoft Fabric, Lakehouse (Medallion Architecture), Neo4j (graph), ChromaDB (vector), PostgreSQL, RBAC
- **Tools:** Weights & Biases (WandB), LangSmith, CVAT, Voxel51 FiftyOne, Automated Testing, DataforSEO API, Protobuf, Git

EXPERIENCE

Systems Limited July 2026 – Present
Senior AI Engineer *Islamabad, Pakistan (On-site)*

- **HR Data Chatbot:** Multi-persona enterprise HR assistant (employee, fine-grained admin, client admin, company-wide admin) offering personal data, **benefits and retirement planning** suggestions, and workforce analytics; built on a **Microsoft Fabric medallion lakehouse**, deployed on **OpenShift**.
- Designed the **knowledge graph ontology** for HR data and built the **ingestion and sync pipeline** from the Fabric gold layer into **Neo4j**, carrying **source-system RBAC** into the graph.
- Built the **LangGraph** AI layer (orchestration and **role-based question answering** scoped to each user's permissions); currently building an **automated test suite** for all use cases across personas.

PyFlowLabs Pvt. Ltd. August 2025 – June 2026
Senior Machine Learning Engineer *Lahore, Pakistan (On-site)*

- **Auto Parts Catalog & DIY Repair AI:** Built a knowledge graph of OEM part **cross-references, successions, and fitment** from extracted spare part standards, a **standards RAG knowledge base**, and an **agentic DIY Repair assistant** for guided diagnosis and repair (LangGraph, Azure OpenAI, Neo4j, ChromaDB).
- **Genie — Privacy-Preserving Visual Recommendation:** Fine-tuned **LLaVA 7B** on **50,000 image-caption pairs** with PEFT; served via **TensorRT** with dynamic batching; designed a **split-inference architecture** so raw images never leave the device; linked entities to a **Wikidata knowledge graph** for recommendations.
- **HR Profession Ontology & Agentic Recruiting:** 3-tier profession taxonomy (Job Family → Occupation → Job Title) with skills linked via a knowledge graph. Classified **1M+ LinkedIn job titles across 20 languages** via LLM pipelines; deployed multi-agent recruiting system (LangGraph, Azure OpenAI, Neo4j, ChromaDB, LangSmith).
- **Amazon SEO & PPC Optimization:** Agentic SEO pipeline (search terms → DataforSEO → embedding clustering → LLM listing generation) and an agentic PPC bid recommender.
- **Odyn — GPU Container Orchestration Platform:** RunPod-style GPU cloud from scratch: Go/gRPC control plane, Go node agents, Docker + NVIDIA containers, FRP tunneling (zero inbound ports), Ray LLM inference routing (OpenAI-compatible API), GitHub Actions CI/CD.

Opthalytics Inc. (USA) January 2025 – July 2025
Deep Learning Engineer *Pakistan (Remote)*

- Trained **lesion detection models** and developed algorithms to predict **diabetic retinopathy (DR)** from **ultra-widefield (UWF)** and standard **fundus** retinal images; wrote the ML backends for model training and inference.
- Built a **medical data platform on Azure**: an orchestration backend that tracks every image, a **deduplication pipeline** for onboarding new data, **labeling and cross-review jobs** for ophthalmologists in self-hosted **CVAT**, and **Voxel51 FiftyOne** for data review and model evaluation.

TheWalnut.ai Pte. Ltd. September 2022 – December 2024
Machine Learning Engineer *Kuala Lumpur, Malaysia (On-site)*

- Achieved **TEDS 98.51 on PubTabNet** and **TEDS 99.45 on FinTabNet** for table structure recognition.
- Fine-tuned **LayoutLMv3 / Florence-2 class** transformers using full fine-tuning and **LoRA/PEFT** on 10^4–10^5 document pages with **PyTorch DDP** and **WandB**; applied **knowledge distillation** to compress document backbones for production.
- Reduced inference latency **5×** via **ONNX** → **TensorRT** (GPU) / **OpenVINO** (CPU), **PagedAttention**, and dynamic batching; built custom mAP/IoU evaluation pipelines.
- Built supervised and self-supervised training pipelines on **AWS, RunPod, Lambda Labs, and DigitalOcean**; deployed model services via **AWS Lambda** and RunPod.

Selected Projects:

- **Vegaspread — Financial Document Intelligence:** OCR → multimodal layout detection → table structure recognition → NLP hierarchy construction → agentic validation → structured output for credit risk assessment.
- **Contract Reader** (summarization + RAG Q&A); **Signature Verification & Fraud Detection PoCs** (GAN image cleaning, one-shot verification, transaction anomaly detection).

VisionLab, SEECS NUST April 2022 – August 2022
Research Assistant *Islamabad, Pakistan (On-site)*

- Contributed to a **government-funded** smart agriculture project; trained deep learning models for automated crop disease detection from drone imagery.

Huaneng Power International June 2015 – March 2022
Mechanical Engineer & Supply Chain Manager *China & Pakistan (On-site)*

- Directed commissioning of a **1,230 MW** supercritical power plant; applied predictive diagnostics and data-driven modeling for supply chain and asset optimization across international operations.

INDEPENDENT PROJECTS

NLLB Low-Resource MT — EdTech (Freelance) *2024*
Fine-tuned Meta's **NLLB** for low-resource **Quechua** ↔ **English** translation using **QLoRA** and full fine-tuning with PyTorch DDP; evaluated with BLEU and chrF.

EDUCATION

MS Artificial Intelligence August 2025
National University of Sciences and Technology (NUST), Islamabad, Pakistan
CGPA: 3.85/4.0 | Paper under review
Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Distributed Systems & Cloud Computing

BS Mechanical Engineering June 2015
CGPA: 3.20/4.0

PUBLICATIONS

- **LEGR: Learnable Edge Graph Refinement for Table Structure Recognition** — submitted to *Artificial Intelligence* (Springer), under review (First Author)

CERTIFICATIONS

- Machine Learning Specialization — Deeplearning.AI / Coursera | Deep Learning Specialization — Deeplearning.AI / Coursera | Statistics Specialization — Coursera